



Metaməlumatlar

SƏNƏD

Ad

ACTIVE LEARNING FOR NAMED ENTITY RECOGNITION: REDUCING LABELED DATA REQUIREMENTS

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Yarımşöbə

MBA-magistratura

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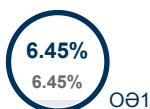
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BAKU - 2026

ACTIVE LEARNING FOR NAMED ENTITY RECOGNITION: REDUCING LABELED DATA REQUIREMENTS

ABSTRACT

Research relevance: The rapid growth of digital platforms and online educational environments has resulted in a substantial increase in unstructured textual data. Extracting meaningful and structured information from such data has become an important challenge in Natural Language Processing (NLP). 58 Named Entity Recognition (NER) plays a critical role in converting unstructured text into structured information that can support analytics, automation, and decision-making processes. However, modern NER systems require large annotated datasets, and the process of data labeling is expensive and time-consuming. Consequently, active learning has emerged as a promising approach for reducing annotation requirements while maintaining model performance. Therefore, investigating the integration of active learning techniques into NER systems represents a relevant and significant research area.

Research objective: 3 The main objective of this study is to develop and evaluate an active learning framework for Named Entity Recognition that minimizes labeling requirements while preserving predictive performance. The study aims to compare active learning with traditional supervised learning approaches, assess annotation efficiency, and analyze how the extracted structured data can support business-oriented insights within the context of online course platforms.

Research methods used: The study applies data preprocessing, tokenization, BIO tagging, feature engineering, and iterative active learning techniques. A BiLSTM-CRF model is implemented as the baseline NER architecture, while uncertainty-based and diversity-based sampling strategies are integrated into the active learning framework. Model performance is 13 evaluated using precision, recall, and F1-score metrics. In addition, visualization and comparative analysis techniques are employed to examine category demand, pricing strategies, course duration, and partner performance within the dataset.

Research database: The research is based on a dataset consisting of 1,000 online courses collected from the FutureLearn platform. The dataset includes textual course descriptions, pricing information, duration, category labels, learner ratings, and enrollment statistics. Academic literature related to Named Entity Recognition, active learning, and Natural Language Processing is also utilized throughout the study.

Research limitations: The study is limited by the use of a single-domain dataset and a controlled experimental environment, which may affect the generalizability of the findings. The proposed framework is evaluated using a BiLSTM-CRF architecture and does not include transformer-based models such as BERT. Furthermore, the business-oriented analysis relies on aggregated platform-level data and does not incorporate user-level behavioral or temporal factors.

Scientific novelty and practical significance of the research: The study contributes to the literature by proposing a structured active learning framework that integrates uncertainty and diversity-based sampling strategies for efficient Named Entity Recognition. The research demonstrates that active learning can significantly reduce annotation requirements while maintaining competitive model performance. In addition, the study extends the practical

application of NER by transforming extracted textual information into business-oriented insights related to demand patterns, pricing optimization, and platform performance. The findings provide practical guidance for organizations seeking cost-efficient NLP solutions and data-driven decision-making approaches.

Keywords: Named Entity Recognition, active learning, Natural Language Processing, machine learning, annotation efficiency, BiLSTM-CRF, online education platforms, text analytics.

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INTRODUCTION

Relevance of the Study: The dramatic increase in the digital platform has led to the unprecedented increases in the volume of unformatted textual information, even in the education sector of the Internet. Websites like FutureLearn offer thousands of lectures that are textually explained about each lectures, which comes with invaluable information regarding it, its giver, cost and interaction with learners. It is important to extract structured knowledge of such data into enhanced decision-making, individualization and optimization of platforms (Hedderich et al., 2021).

Named Entity Recognition has played a crucial role in transforming the unstructured text to the structured text based on the identification and classification of important entities. Nevertheless, with the current developments in deep learning, NER systems still heavily rely on big annotated samples. This requirement limits their applicability in practice within practical real-world contexts where labelled data are expensive and time-consuming to obtain (Lample et al., 2016; Devlin et al., 2019).

A possible solution to this problem is active learning since it enables models to learn using fewer labeled data. The cost of labeling, which when applied to a large sample of the data is very expensive, can be reduced significantly since active learning can selectively apply the most informative samples to be annotated and yet still translates to competitive performance levels (Settles, 2012). Therefore, it will be timely and pertinent to explore the active learning technique in the field domain-related tasks, i. e., information in online courses.

Problem Statement: Modern NER models are effective but they greatly require the existence of large, high quality labelled datasets. In practice this is a labor-intensive task to have such datasets, particularly when annotating on a token-level is needed, and when working with domain-specific entities. This poses a serious impediment to the implementation of the NER systems within the actual environments such as online learning platforms (Tjong Kim Sang & De Meulder, 2003).

The textual description in course data surrounds a multifaceted and rich information, e.g., names of the institutions, course price, time, and course classification. It is not an information that can be easily and conveniently annotated manually. Next, strategies that can be adopted to reduce the number of annotations needed without compromising the performance on the model is of interest.

A potential solution is active learning since it pays attention to annotating the most informative samples. However, there is not much information on its performance in domain-specific NER tasks, particularly using the data of online education. The current paper fills this gap by modeling and experimenting an active learning model in this area..

Motivation for Active Learning Approaches: Traditionally, the construction of high-performance systems based on Named Entity Recognition has been built on the large-scale manually-marked data. Nonetheless, in the analysis above, we can find that data annotation is not only costly, but time consuming in addition to the fact that it may involve domain knowledge. Such constraints bring a lot of difficulty especially in specialisation areas or languages that do not have a lot of resources where the labelled data sets are rather sparse (Settles, 2012). As a result, the desire to hold paradigms of learning which lessen the degree of annotation yet are still able to be competitively significant handles an increasing purpose.

Active learning offers a philosophical approach towards ensuring that this is taken care of. The active learning instead of the random selection of data to be labeled can give the model the capability to learn and demand to be labeled the most informative samples. The model may be utilized to generate a more effective performance than the traditionality of the supervised training by considering the cases of uncertainty, diversification, or the representative

cases (Lewis & Gale, 1994). This discriminative annotation scheme is especially appealing to sequence labeling tasks like NER because the cost/instance of annotation is high in this case.

Furthermore, contemporary deep learning models as a training model usually need a large dataset to effectively work well. Nonetheless, there are also some empirical indicators according to which not every data is equally useful to the learning process. Others have samples the contribution of which to the model on acquiring the common patterns is negligible. Active learning exploits this and concentrates effort on annotation of data which encourages the maximal learning. This on top of saving the number of hours and efforts that were spent in labeling would lead to a higher rate in the modeling development cycles.

The additional triggers to active learning is accompanied with dynamic environment in which new forms of entity change or domain changes are constituted. The model uncertainty aspect on pubertal annotation provides the opportunity to keep on improving the system without having the anxiety to re-annotate the information. Through this, there will be an effective and more flexible active learning that is the sunny side of marketing NER systems with a small labeling budget.

Research Objectives and Questions: The aim of this study is to investigate the extent to which active learning strategies can reduce labeled data requirements in Named Entity Recognition (NER) while maintaining competitive predictive performance. To achieve this, the research focuses on the design, implementation, and evaluation of an active learning framework integrated with a neural NER model.

The objectives of the study are:

1. Develop a baseline supervised NER model to serve as a performance benchmark
2. Design and implement active learning query strategies for selective data annotation
3. Evaluate the impact of active learning on annotation efficiency
4. Assess model performance under varying labeling budget constraints

To guide the investigation, the following research questions are formulated:

1. To what extent can active learning reduce the volume of labeled data required for effective NER performance?
2. Which query strategies are most effective in improving model performance under limited annotation conditions?
3. How does model performance evolve across successive active learning iterations?
4. What trade-offs emerge between annotation cost and predictive accuracy?

By addressing these questions, the study seeks to contribute both theoretical insights and practical guidance on the application of active learning techniques in NER tasks.

Scope of the Study: This study focuses on the application of active learning within the context of Named Entity Recognition using a supervised deep learning framework. The research evaluates the effectiveness of selected active learning strategies in reducing annotation requirements compared to traditional fully supervised approaches.

The study is conducted within a controlled experimental environment, utilizing a neural NER model integrated with iterative active learning cycles. Performance is assessed under varying labeling budgets to determine the efficiency of the proposed approach. The analysis is limited to standard entity types and a structured dataset, allowing for consistent and controlled evaluation conditions.

It is important to note that this study does not aim to provide a comprehensive comparison of all existing NER architectures or active learning algorithms. Instead, it focuses on demonstrating the practical benefits of active learning within a defined experimental framework.

Furthermore, the scope excludes large-scale multilingual applications, real-time production system integration, and extensive cross-domain generalization. These areas are identified as potential directions for future research. The primary emphasis remains on validating the effectiveness of active learning in improving annotation efficiency while maintaining model performance.

Research Method Overview: The research design used in this paper is an experimental one with machine learning and analysis. To aid in the activity of NER, online course description dataset is collected and preprocessed. It provides a minimum NER model, following the architecture of BiLSTM-CRF, and is trained on a very small size data set of labeled data.

This is followed by an active learning model where by the model repeatedly selects informative samples to be annotated or not based on criteria of uncertainty and the level of diversity. This model is retrained at the end of each iteration and performance is measured on standard measures: precision, recall, F1-score.

In addition, the extracted entities are analyzed to make business-oriented analysis supported by visualizations, which demonstrate trendy and meaningful relationships in the data. The unified practice allows technical analysis and the creation of practice.

The research is based on an experimental approach for data mining and machine learning. A Named Entity Recognition (NER) model is proposed and combined with an active learning strategy to minimise the cost of labelling. It starts with a model trained on a small, labelled sample, and then selects samples for labelling based on uncertainty and diversity. These are labelled and added to the training set. Then it is evaluated against the usual measures and the labelled data are used for business purposes.

Structure of the Thesis: The thesis has three chapters. Chapter 1 introduces the background of Named Entity Recognition and active learning, and the concept model. Chapter 2 describes the method, including data gathering and preparation, model preparation and active learning. Chapter 3 covers the result of the model performance evaluation, business view analysis of the extracted information and discussion. The thesis concludes with conclusions, limitations and future work, references and appendices.

12. Limitations : This study is subject to several limitations. First, the analysis is conducted on a single dataset derived from an online learning platform, which may limit the generalizability of the findings to other domains or datasets. Second, the experimental setup is implemented in a controlled environment and does not fully capture real-world deployment challenges such as annotation inconsistencies, scalability issues, and dynamic data changes. Third, the NER model is based on a BiLSTM-CRF architecture, and more recent transformer-based approaches are not explored, which may influence performance comparisons. Finally, the business analysis relies on aggregated data and does not incorporate user-level behavioral factors or external variables that may affect learner engagement. These limitations are acknowledged and provide directions for future research.

CHAPTER 1. NAMED ENTITY RECOGNITION

1.1 Foundations and actuality of Named Entity Recognition

NER is one of the fundamental natural language processing (NLP) tasks, which deals with the recognition and categorization of named entities, i.e. person names, organizations, places, date and many others, into already known categories in an unstructured text. NER is a major component of a variety of downstream NLP applications (such question answering, information retrieval, knowledge graph construction, machine translation, or text summarization), so it is one of the earliest information extraction tasks (Hu, 2024). The named entity detection with high precision can be used to identify

useful organization in the written data and can be operated to carry out the reactive operations on the large corpora.

The early NER systems were quite rule-based and expertly created lexicon. Such systems that were manually programmed produced linguistic rules, regular expressions and gazetteers to find entity spans. The rule-based methods have shown to be sufficient in the small and limited scope but could not be agile and sizable in respect to wide linguistic contexts (Jehangir et al., 2023). The inflexibility of homemade rules did not allow them to be very flexible to new worlds and languages and maintaining such systems required a lot of manual labor.

The switch to the systems relying on the use of the statistical machine learning was an impressive addition to the NER research. NER can be viewed as a probabilistic sequence labelling task, and can be solved with sequence labelling models, like Hidden Markov Models (HMMs) or Conditional random fields (CRFs). The models are contextual dependencies central on the token and engineered attributes as part-of-speech tags, orthographic pattern and word window. The emergence of CRF-based models and especially, their capacity to capture sequential dependencies gained popularity because the former generative models were unable to do the same (Jehangir et al., 2023).

Deep learning has appeared and this caused some drastic change in the sphere of NER. ANNs like Bidirectional Long Short-Term Memory networks (BiLSTMs) that can be stacked with CRF decoding layers minimized the need to use designed features as they are learnt to obtain a representation of contexts on a data basis. This is another manner in which word embeddings were also beneficial in enhancing performance by offering the dense semantic representation of the tokens (Hu, 2024). Multimodal neural methods were more generalized and had a high-performance in case of global benchmark data testing.

Recently, Transformer-based pretrained language models (PLM), such as the ones called BERT, and their variants have been again reinvented as NER. The postulates of these models are unsupervised pretraining of a large scale of written corpus that are of enormous scale to acquire contextual representations of words that can not only acquire syntactic but also semantic qualities. Such models on refinements on NER activities have shown a great accuracy, recall, and F1 score improvement on various information (Recent Advances in NER Survey, 2024). The transformer based architectures are better in the aspects of long range dependence and contextual subtleties which is paramount in the eradication of loose entity limits and forms of entities.

The contemporary NER systems have been becoming multifaceted despite the remaining enhancement owing to the changing demands. Embedded entities, domain specific terminologies, polydimensional text and dynamically varying entity inventories tend to be identified to be applied to real world. As an example, biomedical NER will be compelled to acknowledge highly specific objects, e.g the names of genes and chemical substances, and the legal NER must acknowledge laws and references to cases. They are domain related problems, which require domain specific data sets and domain adaptation (Jehangir et al., 2023).

Besides, explanatory power, as well as interpretability have acquired the issues of study in recent time. Since NER systems are deployed in resourceful practices, like medications and finance release, the issue of what models do and why they fail is a necessity to understand. The investigation of the mechanisms of attention, inquiry techniques, and approaches to analyzing the mistakes becomes usual to explore the reflection of the information about the entities on the neural models (Hu, 2024).

History Named Entity Recognition (NER) systems have over the years been gradually upgraded with the rule based models being converted to external data-driven statistical algorithms. This characteristic is reflective of the general literature in the field of Natural Language Processing (NLP) in which preceding the development of the machine-based learning algorithms capable of learning a pattern by noticing data, a significant amount of work occurred in the field which were based upon the linguistic rules and information gathered manually.

Existing NER systems Early NER systems that existed were rule based and depended on weak ad hoc written linguistic rules, pattern matching and domain specific vocabularies also known as gazetteers. It was the experience in the field that resulted in the creation of such systems; the basic knowledge of the language structure, morphology, and semantics turned into a number of rules that could assist to define boundaries of entities and the type of entities (Grishman, 1995).

The rule-based systems normally handle a combination of regular expressions and looked ups. In the case of the capitalization patterns, named entities are identified by use of contextual cues (e.g. titles, e.g. Dr. or Mr.) and known lists. Among the most true corpus of this process are the gazetteers, the lists of the names of persons, associations, and names of places, edited.

The benefit of the rule-styled systems is that the system is very precise in the managed scenario. Such systems have been in position to provide correct output in particular areas where there is predictability and limited trends of entities. The rule-based approaches can as well be viewed as the method, which incorporates the decision-making procedure, as established in rules that can be understood by a human being (Nadeau and Sekine, 2007). Nevertheless, there are also many pitiable drawbacks regarding rule-based NER systems. To begin with, they are not scalable because it is difficult and time-consuming to develop and keep the rules of areas and other languages. Second, they have low levels of generalization in their inability to cope with new or undetermined appearance of things. Third, the rule based systems are very susceptible to linguistic variations and thus, this renders it hard to modify them to align with the new linguistic practices (Poibeau and Kosseim, 2001). Consequently, to the extent that rule based systems were the pioneers of early research in the topic of NER, it cannot be utilized with efficiency on bigger and dynamic scales.

The shortcomings of the rule based techniques were beaten through statistical techniques thereby providing a method that is more scalable and will be practical. These system solve NER as a task involving supervised learning and learners are trained on annotated data to learn concepts to be able to distinguish between entity types and non-entity tokens. Generative models (Hidden Markov Models(HMMs)). were one of the first statistical NER approaches. It is the nature of the HMM-based systems to model the joint probability of the following series of observations (words) and the hidden space (labels on entities) and the ability to predict the sequence of labels which is likely to appear in case the text is presented as an input (Bikel et al., 1999). Although HMMs provide a probabilistic modeling of sequence labeling, HMMs are very strong bearing in mind that they are also better shaped to take drastic independence assumptions that restrict their model of formalizing the complex context dependency.

Introduction of Conditional Random Fields (CRFs) was the other important development that was specifically-designed discriminative models used in sequence labeling. CRF is not the only model contrary to HMMs, as there are a lot of features incorporated in it, being extremely numerous and not independent to one another, which can be reflected as the conditional probability of label sequence, under the conditions of considering an input sequence (McCallum and Li, 2003).

The suitability of the model to calculate contextual dependencies and a constraint of sequence consistency between labelling sequences has been said to explain the rapid evolution of CRFs in NER. It is explained by the fact that CRFs may test it to make sure that there is no example of an Inside (I) tag that lacks a supporting Beginning (B) tag so that there are still legal entity structures to be encountered. In addition, CRFs enable more sets of complex features such as lexical features (identity of a word), orthographic features (capitalization, digits) and context features (the presence of neighboring words and tags).

Another significant statistical NER model is the Maximum Entropy Markov Model (MEMM) a combination of maximal entropy classifier, as well as sequential modeling. Nevertheless, MEMMs are biased in the labeling such that propensity towards states with less outgoing transitions and thus are not

as efficient as CRFs (Lafferty et al., 2001). Feature engineering plays a central role in statistical NER systems. Researchers have developed extensive feature templates to capture linguistic patterns, including:

- 1. Word-level features (e.g., current token, previous/next tokens)
- 2. Character-level features (prefixes, suffixes)
- 3. Part-of-speech tags
- 4. Gazetteer matches
- 5. Word shape patterns (e.g., "Xxxx" for capitalized words)

The good quality and relevance of these features to the quality of statistical models is a great one. The adherence to documentation of feature sets by the designers should be considered because it can cause the NER performance to skyrocket, this fact presupposes that the role of the domain knowledge should be given particular attention irrespective of the data-driven paradigm (Ratinov and Roth, 2009).

The reason is also that statistical NER systems are arguably flawed despite their associated benefits. To begin with, they need to access considerably large volumes of tagged training data that may be expensive and time tested in order to obtain. Second, the cost of feature engineering is excessive and in certain instances applicability of the models to different applications. Third, statistical models can be affected by long-range dependencies of a text, in which computers are typically trained to think in square window frames.

Limitations of the rule systems and also of the statistical systems have led to development of the neural network-based methods which automatically learn features and have higher chance of identifying complex situational relationships. It needs to be mentioned that most of the current NER systems are yet to be based on the aspects of the former strategies. A case in point is the normative use of gazetteers as an aiding bit of hybrid model and CRF layers are commonplace an element of decoding within a neural network. This emphasizes the fact that rule-based and statistical systems have always remained default components of NER systems

1.2 Challenges and Evaluation in Named Entity Recognition

The paradigm of training an NER system of high-performance is the existing supervised learning paradigm. The accessibility of large and well-marked data however is a major concern in respect to the success of the supervised models. Such datasets require both time and money to create, not to mention that they are usually domain-specific, an even more serious constraint as far as NLP research is concerned, not to mention the use of this method in the industry (Jehangir et al., 2023).

High cost of manual annotation is one of the big problems. ERER in comparison with basic text classification tasks involves token level marking with accurate spans and entity types. As the annotators themselves are keen on assigning the point of beginning and end of an entity, it is an infixed case in case of a nested or overlapping entity. It requires skills in languages and knowledge of regulations in annotation. There are parts of the domain knowledge to be needed, like the field of medicine or law, which can further make the cost of the annotation even higher (Horych et al., 2025). As such, ensuring a high quality and large scale NER is an expensive endeavor in comparison with the amount of finances and even human capital needed to accomplish it.

Imbalance and deficit of data is another important problem. The real-world corpora have skewed distributions which are characterized by a small number of common types of entities being much more common and other types which can be domain specific entities being much rarer. The distorted data is also the reason why the model is being trained well with the common entities and poorly with the rare ones. This distribution of a long-tail is discouraging the generalization of such a potential issue, and it decreases reliability of a given system in practice (Hu, 2024). Also, the specific categories of rare items would be those that offer the greatest value in the activity of the expertise, e.g. rare diseases in the medical records or new company names in the stock market news.

Another problem is the domain adaptation. Language models that are typically pretrained are predicated on general-domain corpora, e.g. web text and books. Application to special fields might drag down on performance by use of vocabulary and context change. In addition, the schema of annotations of data is different. There is simple flat entity structure and others may be of nested entity structure or hierarchy. The differences render transfer learning to be challenging, and reduce the reusability of labelled data (Recent Advances in NER Survey, 2024). Due to it, organizations might have to regularly build new domain annotated datasets by making more and more labeling.

Major problems with the quality of annotation and inter- rater agreement are also present. Even those who are annotators that are hired by the professionals themselves might not always be in unison when it comes to the scope of the entities or the terms used to identify types especially when the situation might be uncertain. These contradictions add noises to labels and it may have a negative impact on the model training and evaluation process. To achieve the goal of making sure that there was congruency in annotation, repeated annotation, evaluation and revision of the guidelines brings it to a higher price and increased time durability to release the information (Horych et al., 2025).

Another flaming problem is scalability. Progress in the evolution of the development of static labeled datasets is quick with the consideration that new entities are incorporated on a regular basis in the dynamic environment. Considering the fact that such new terminology or even the speech to the new geopolitical organization may not even be mentioned in the latest commentaries due to the newly formed body or organizations. Sized continuous manual labeling of databases cannot be performed. This is the issue that encourages the adoption of the strategy that can be used to reduce the process of labeling, and prevent the loss of performance (Li et al., 2024).

Such sparse-labeled data evaluation also results in the problem of methodological complications. The small size of the data sets contributes to the variability of the performance measures, as well as decreases the statistical validity. The comparison of models does not have as high sensitivity of use and the model hyperparameter optimization also overfitting. Naturally, more attention is paid to the techniques of the careful validation by researchers and the division of several random groups of observations to reduce the mentioned risks (Li et al., 2024).

According to the current studies, active and weak supervision, and semi-supervised learning can be an alternative to the limited number of labelled data. A particularly competitive performance is sought by semi-mechanical selection of most informative (that is to be annotated) examples performed by an active learning; thus there is a smaller number of labeled examples performed on a total (Li et al., 2024). According to the empirical results, an established active learning models can reduce the annotation fees found to substantially reduce the F1 scores when compared to the fully supervised ones.

Finally, although performance of neural architectures has been significantly enhanced due to the progress made in this area, it must be noted that the fact that the usage of NER systems heavily depends on huge and good-quality labeled datasets is mutually limiting the usage of the systems on low resource or other specialty areas. The new training method is the primary objective of active learning methods studied in the present paper, which is the required solution around which the maximum effectiveness of annotation and learning of few labeled data will be attained.

Table 1. Key Challenges of Labeled Data in NER

Challenge	Brief Description
High Annotation Cost	Token-level labeling requires precise boundaries and domain expertise, making annotation expensive and time-consuming

(Jehangir et al., 2023; Horych et al., 2025).

Data Imbalance Real-world datasets are skewed, with rare entities underrepresented, reducing model generalization (Hu, 2024).

Domain Adaptation Models trained on general data struggle with domain-specific vocabulary and structures (Recent Advances in NER Survey, 2024).

Annotation Inconsistency Annotators may disagree on labels and boundaries, introducing noise into training data (Horych et al., 2025).

Scalability Constant emergence of new entities makes static labeled datasets quickly outdated (Li et al., 2024).

Small Data Issues Limited data leads to unstable evaluation and higher risk of overfitting (Li et al., 2024).

Data Dependency NER models require large labeled datasets to perform effectively (Jehangir et al., 2023).

Need for Efficient Methods Traditional supervised learning is inefficient under labeling constraints, motivating active learning (Li et al., 2024).

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NERs evaluation is the mandatory process of the model performance assessment, particularly, in the case of the tasks related to the sequence labeling which is rather difficult. In contrast with the usual classification predicaments, NER does not only demand adequate classification of the entities, but also a precise location of the entity delimiters within the text. In that regard there were certain steps of evaluation that needed to be formulated, which would provide an idea of the accuracy and the level of comprehension of the entity recognition. Precision, F1-score, and recall are the most popular tools of measurement in the NER. The information retrieval provides the basis of these metrics which give the complementary views of the model performance.

²⁹ Precision measures the proportion of predicted entities that are correct:

²⁹ Recall measures the proportion of actual entities that are correctly identified:

²⁹ F1-score is the harmonic mean of precision and recall, providing a balanced measure of performance:

The measures are particularly crucial to NER in light of the fact that high level of precision can be realized by a model by predicting an excessively small number of entities by the model or high recall can be realized by predicting a very large number of entities through a model, but a trade off between the two can be the most effective indicator of a good performance (Sokolova and Lapalme, 2009).

There are varying levels of granularity in which the NER performance can be measured, the most self-evident of which is the token-level performance and the entity-level performance. In token-level evaluation, an evaluation is conducted on the word level and whether the word was a match of the word predicted or not and the probability of it being a match. Although it is an easy way, it is also deceptive because it fails to investigate the correctness of complete entity spans. As an illustration, one may observe that though the entity boundaries were specified wrongly, partially correct predictions may be referred as strongly correct at token level. Conversely, entity level assessment exists at the boundary level, whereby the presence of the specific matching predicted and actual entities in terms of boundary and labels exists. An entity that is predicted to be correct must have all three attributes to the position, the end position and the type, which are ground truth value. This increases the entity level assessment and is closer to the actual performance in the real life (Tjong Kim Sang & De Meulder, 2003). One of the most popular evaluation parameters of most modern NER tasks, and the CoNLL-2003 shared task is entity-level F1-score.

Among the issues related to the NER assessment is the query of dealing with a partial match. Strict evaluation surroundings constitute those where only the accurate responses can be discussed as precise i.e. any differences between them and the entity boundaries or type are referred to as false prediction. Part matches can however, be informative. Indicatively, the knowledge of New York as opposed to New York City is not yet none worth wanting, but technically incorrect as would easily have been thought of in terms of the rigorousness of judgement. In order to provide an answer to this aspect of the analysis, a number of studies provide a set of slack evaluation measures, which take partial coincidence concerning the predicted and true entities. Such methods will give more insight into the model performance especially when it comes to situations that are compound and involves worrying about nest or blurry when one is keen on things (Batista et al., 2018).

Speaking of the NER system performance concerning different types of organizations, one may micro-average or macro-average it. Micro-averaging is a global set of calculations on the measures which sums the sum of the true positives, the false positives and the false negatives of all the types of entities. The common types of entities receive a much heavier emphasis in such approach, and this is the one that is typically employed in NER benchmarks.

Macro-averaging in its turn calculates separately the metrics by the type of entity and averages them. It equalizes all types of entities irrespective of how common they are and offers the interpolation of the performance of the line that is rarely displayed or underrepresented (Sokolova & Lapalme, 2009). Such techniques are subject to the aim of the study. Micro averaging would help in optimizing the overall performance of the system yet macro averaging would be important in the process of researching on the inherent unbalance in the classes.

The evaluation of the situation when there is limited labelled data such as active learning is more complicated. The smaller data sets may result in higher variance of performance measures resulting in less stable/ stable performance and sensitivity of splits of data. In a bid to overcome this, the researchers have been keen on cross-validation techniques, robustness of experiments, learning curves, to achieve good assessment. The learning curves can constitute the active learning research because the curve is used to show how the model performance changes as more amounts of labeled data are presented (Settles, 2012). Performance/annotation cost relationship is another parameter that is supposed to be taken seriously. The active learning in active learning is not aimed at maximising F1-score, but high performance, with less labeling effort. Most cases, hence, cost-efficiency tools, including performance gain per annotated sample, have been included in the evaluation system.

What is very important in NER studies are test measures and normalized data. Annotated corpora Academic corpora like CoNLL-2003, OntoNotes, etc. include an error training, validation and test split of the corpus. These criteria make it possible to make a conventional comparison between various models and approaches. These datasets are accompanied by assessment scripts that make sure that measurements are identified in a predictable way to minimize the possibility of variations between the reports that are given twice. CoNLL analysis program is one example and mandatory flag of entity level matching and it is an accepted standard in the community (Tjong Kim Sang and De Meulder, 2003).

1.3 Active Learning Approaches in Natural Language Processing

The basic learning process is query strategies which entails active learning which determines the points of data to be annotated and hence the sample size of annotated data employed to learn a model in a cost-effective manner. One of the most crucial factors which define the informativeness and representativeness of the selected samples is quality of active learning system query strategy since the effectiveness of this part is directly influenced by this aspect. Contextually, in Natural Language Processing (NLP) and more importantly, in a problem such as Named Entity Recognition (NER), query strategies need to take into account the complexity of the language and the active influence of the context, ambiguity and feature underlying-dimensionality. The existence of this fact has led to the development of a large number of query strategies all of which are aimed at the informativeness of different dimensions and improving the level of model performance in an environment when annotation budgets are low (Settles, 2012).

One of the most commonly used approaches of the active-learning is uncertainty-based sampling, which has its ground on the premise that the model in question is expected to focus on those cases in which it is most inaccurate in its predictions. The rationale of this is that non-certain samples have higher probabilities of providing new information that will reduce the decision frontiers of the model. The uncertainty can be quantified in numerous ways, they are least confidence, margin sampling and entropy based measure. The fact that it was so unclear on the foundation of entropy that is utilized in information theory is particularly effective in that it represents all of the probability distribution of potential labels, and not the most significant predictions (Shannon, 1948).

Trying to eliminate the negative aspects of single-model estimation of uncertainty, the query-by-committee (QBC) approach suggests a new approach to uncertainty by the means of discord of several models. A panel of models which have been trained on the same marked dataset are typical here usually via initializations or parts of the information. These degrees of difference in the forecasts of the committee members are then engaged as a measure of feel of uncertainty. These cases that create extensive disagreement are considered to be more informative and are annotated. This approach is particularly rational with respect to epistemic uncertainty which is brought about by ignorance of the model on certain areas of the data space (Seung et al., 1992). By reducing the bias of a model, which is an inherent weakness of NLP applications, QBC can also increase the strength of these applications and allow more balanced learning. Cost of increased computation is however large because of the need to possess multiple models therefore, this methodology is not as feasible as large scale or factor constrained applications.

The other significant type of query strategies is one that focuses on diversity based sampling that is supposed to be non-biased when selecting samples that would be representative of the entire data set. Even though uncertainty-based approaches put a high value on informativeness, they are still subject to yielding a biased choice of similar/redundant samples when dealing with data sets in which patterns are clustered closely. These effects are mitigated by the diversity-based techniques which select samples that are not similar to one another and thus spread to a broader part of the input space. This will either be through clustering, distance measures in embedding space or by optimization methodologies such as the selection of core-sets. The idea of diversity is generally measured in relation to the representations of features calculated at the intermediate levels of neural networks in deep learning, which also enables making more extensive comparisons of points of data (Sener and Savarese, 2018). It is also useful in NER tasks especially in order to contain the diversity of the type of participants (i.e., the variations in the type of entities), contexts, and linguistic patterns that are critical in building powerful models.

It has become understood that no individual strategy can exist on its own and subsequently, more studies carried out in the recent past have been focused on hybrid query strategies that represent the composite of differing criteria, such as uncertainty and diversity. The aims of these strategies are to trade-off between exploration and exploitation by sampling very informative and representative samples of the data distribution. An example of a hybrid strategy would be to first identify a collection of uncertain samples and a diversity filter which will be resorted to to filter such samples. The rate of selection is 2-staged and has a high rate of information and prevention of redundancy. Hybrid approaches are also shown in particular to be useful in deep learning where the models learn in high-dimensional space of features and that uncertainty and diversity are both essential elements that guide learning (Ash et al., 2020).

Along with these popular methods, theoretically-based query methods have been proposed, such as expected model change and expected error reduction. These procedures also make a direct estimate of how the labelling of the specific sample would affect the future behavior of the model. It is anticipated that model change maximizes the number of such instances that would lead to the maximum reduction of model parameters, and error reduction through expected improvement attempts to control the amount of generalization error in a model. The idea of having such strategies is countered by the fact they are computationally expensive and difficult to implement in practice on complex information processing models such as deep neural networks (Roy and McCallum, 2001).

Deep learning has demonstrated novel challenges and opportunities of query strategy design. Neural models particularly those based on transformer architecture work in high-dimensional spaces, and most of them do not give well-calibrated uncertainty measures. In more recent techniques, such approaches as Monte Carlo dropout have been applied to the estimation of the uncertainty in the Bayesian sense, and thus uncertainty can be more accurately approximated in a deep model (Gal & Ghahramani, 2016). Other methods that make use of information depending on gradients to estimate the effect of individual samples on the update of the model are a more direct method to measure informativeness (Ash et al., 2020). These techniques are also applicable in the sense of NER wherein, structured format of sequence labeling is a strict condition as one has to select entire sentences and not single tokens in preserving a contextual integrity.

There are two broad categories of active learning structures insofar as the data access that is not labelled as well as query determination methodology are concerned. Two of the most common paradigms would be the pool-based and the stream-based active learning. The fundamental assumptions of these techniques lie widely apart in terms of cost of computerisation, cost of data to be used and their feasibility and application. The differences are the important to consider within the framework in order to acquire the suitable framework to be utilized, especially the NLP issues like the Named Entity Recognition (NER) where data format or the high cost of the annotation are one of the central issues.

The common paradigm embraced in the NLP studies has been the active learning in the pools. The first case, in this case is the fact that it will be a mountain of unlabeled data and the model will arbitrarily choose a few samples within it to label them. The model analyzes all the unlabeled samples or a random subset of them during the i th iteration, according to a query plan and ranks the samples by their instructiveness. The most informative samples are then selected that is being marked with an oracle and they are disregarded in the training collection. This will be maintained until one of the stopping conditions is achieved that can be either a desired level of performance or annotated budget (Settles, 2012).

This is the key benefit associated with pool based active learning that it can become flexible and effective. Since model has complete unlabeled bandwidth, it is able to make justified choices when comparing all of the available ones. This can be used to determine informative sample with a greater precision and can be used by the researcher to utilize the application of enhanced query strategy such as uncertainty-based strategy, diversity-based strategy, and hybrid strategy. The strategies provided by pool based approaches are especially appropriate in the framework of NER due to the convenient access of large portion of unlabeled text and requirement of careful judgment of the situation the samples concerned as entire sentences or even documents. Besides this, pool based methods facilitate batch selection and a large number of samples may be drawn on each iteration which leads to effective annotation in a real world (Olsson, 2009).

Although it has its benefits there are also some challenges associated with the pool-based active learning. The computational cost of large dataset of samples may be high due to the computational cost incurred in computing a large dataset they need to be evaluated to determine these, especially with more complicated models like deep neural networks. In order to address this problem, investigators typically use part of the following manipulations: sub-population sampling, scoring approximation, and pre-filtering techniques in order to cut down the cost in computations. Moreover, the fact that unlabeled data is prior knowledge is not always applicable to the dynamic environment that is being produced at a continuous rate and therefore the pool-based approaches might not cut across the board with regards to the running of real-time systems.

The stream based active learning is then contrasted on another premise the points of the data flows one another and the model must make a decision in

real-time regarding whether it should pose a question whether the label of the next point incoming is known. In this regard, the model would measure the data points of the individual, and a form of selection would be applied in an attempt to come up with the data point which would be adequate enough that it would be informative that it would be worth annotating or not. Once the criterion is achieved, the instance is sent to the oracle that labels the instance, otherwise, it is deleted or just executed but not labeled (Cohn et al., 1994).

Nonetheless active learning which depends on streams is no exception and has its own share of limitations. The model does not have the worldview of pool-based techniques as it implies the analysis of the cases individually. This may lead to poor choice of selection decisions that the model was not able to make comparisons of relative informativeness of the samples. Moreover, the stream-based strategies might have a select threshold that is special to avoid making annotation more expensive, and on the same hand, too aggressive query will not assist to locate useful information since an overly conservative query will cause overlabeling (Žliobaitė et al., 2014).

Similar paradigms are given in the membership query synthesis, where the user is asked to give the model instances where the model is supposed to generate synthetic ones and ask its labels. Its theoretical advantage is definite but this approach is hardly applied to the NLP since it is extremely difficult to produce meaningful and grammatically correct text that can be annotated by people with high accuracy (Settles, 2012). Consequently, a stream based and a pool based pool model are extensively used in developing active learning that is applicable in NLP.

Active learning is an area of importance trend in Natural Language Processing (NLP) that has recently been brought about specifically in the problem of limited labelled data. NLP activities assume a lot of annotation which might have to be done by a human who may be able to read between the linguistic lines, the meaning and semantic terms in the region. Thus, there is a high interest in applying active learning to NLP as an assessment of the enhancement of the working with the models to minimize the cost of the annotations. With the active learning, which filters the most informative data points, one can do better use of the available resources and faster develop NLP systems with the high level performance (Settles, 2012).

Among the most striking forms of applications of the active learning concept the text classification their goal is to present the preset textual messages to preset classes. By then the first study by Lewis and Gale (1994) had already determined that active learning may also play a crucial role in the process of minimizing the number of labeled examples that many folds need to have useful classifiers. Ambiguous or untyped samples containing a relatively small number of cases taken of the specific cases can also be easily used to make some changes to the decision borders of models. The technique has been extensively used in sentiment analysis, systems to detect spam messages and classify messages by topic, in which the volume of untyped text can be easily obtained, but which cannot be easily labelled by humans.

Another aspect that active learning has been successfully applied to is the Named Entity Recognition (NER) that once again is directly related to the research. Object recognition and classification in text is an operation that needs to annotate tokens in detail, which is referred to as NER. One of the key strengths of the active learning is the minimization of the annotation since the process of entity boundaries and category annotation is energy-consuming and cognitively overcharged, which is quite time-consuming. The active learning method has been shown to be competitive NER performance with small quantities of labeled examples compared to the classical supervised methods. An example of that can be seen in the fact that Shen et al. (2018) validate that the accuracy of the deep active learning approaches is accurate in high rates in the instances when the uncertainty-based and diversity-based approaches are combined.

A more complicated NLP composite like machine translation has been accomplished with active learning where it is utilized to rank informative couple of sentences to be corrected or annotated. In that aspect, obstacles will be given priority in active learning within the pattern of translation of the hard to be translated sentence and subsequently enhance the quality of translation with minimal number of human resource. On the same note, active learning has enhanced queries exploration and question answering, in which under such a paradigm can be oriented to ambiguous or challenging queries to boost its accuracy, and also enhance its robustness (Zhang et al., 2022).

The active learning model combined with the deep learning model has been providing a new era of possibilities to the NLP applications within the last several years. Transformers and other constructions utilizing the principle of transformers, including the BERT and its variations have been developed up to the level of the state-of-the-art performance in many tasks. These models are more likely to need massive datasets that it can refine on, which may be a viable limitation. Active learning mitigates that limitation in that, it is more effective in the process of data selection in that the models can perform highly with a low quantity of labeled data. It is determined that an active learning and the pretrained language models can achieve significant cost reduction of the annotation without deteriorating the competitive outcomes (Ein-Dor et al., 2020).

It is not possible that NLP is a process of crossing the river blindly to gain the benefits of active learning. The problem is that it is highly challenging to measure the uncertainty of deep neural models. Nonetheless, as opposed to the classical probabilistic models, neural networks often make overconfident predictions, and also can better diminish the utility of uncertainty-sensitive query strategies. In order to beat this, certain researchers have suggested some of the methods to be used to accomplish this include Bayesian neural networks and Monte Carlo dropout as among the methods of making more realistic uncertainty inferences (Gal & Ghahramani, 2016). The other difficulty is huge volumes of data to assess which are not marked are challenging to calculate.

Also, the NLP tasks are more likely to involve the structured representation, i.e. sequences or trees and the informative samples cannot be chosen easily. In NER, e.g. not only the selection of a token is sufficiently challenging since the model expects context of complete sentences so that it can positively predict. This will be through decision of entire sequences that will make queries strategies more complex and to annotation. Moreover, class imbalances and limited appearance of the objects can complicate active learning, as well, because the model can have issues with capturing desirable samples to the underrepresented classes (Settles, 2012).

The other problem which must be considered is that human annotators must be included in the active process of learning. Although active learning will decrease the annotations that will be made, the emphasis that will be put will be the manner in which the samples will be selected. It is possible to expect that the annotators will be able to make the correct labeling of the complex and, potentially, ambiguous instances and an annotation discrepancy may have a certain negative effect on the model performance. The strategies to be accordingly incorporated in the systems of effective active learning should include the quality of annotation like clear guideline, training process and validation.

There had been a great potential in the low-resource and domain-constrained environments in the applications of active learning. An example of how biomedical NLP is using active learning may also be presented like generating high-quality datasets that involve activities like entity recognition and relation extraction of data that are few and very costly to generate when annotated data is only a few and very expensive to generate. On the same note, the problem of the absence of labelled data in the less popular languages may be solved by using active learning on multilingual NLP to encourage classification of the most numerous samples during the annotation (Hedderich et al., 2021).

1.4 Proposed Conceptual Framework for Active Learning in Named Entity Recognition

Our literature review reveals that although significant progress has been made in Named Entity Recognition (NER), state-of-the-art methods still rely on a large amount of labeled data. This is a significant problem in low-resource settings and domain-specific tasks where acquiring labels may be costly, time-consuming and difficult (Hedderich et al., 2021). Several active learning methods have been proposed to reduce labeling costs but much of this work is

specific to a particular query strategy and/or data set, rather than offering an overview of how to improve labeling efficiency systematically.

To overcome these problems, this paper offers an abstract framework of integrating active learning into the NER framework to allow efficient learning with minimal labeling while maintaining quality. Our framework assumes limited data, and integrates approaches from the literature into a whole and iterative process. The innovation is the integration of different methods (mostly uncertainty- or diversity-based sampling) into an active learning framework for learning, rather than a new method.

Our framework has several components. First, human labels a small amount of data (also known as the seed set). This is used to train the initial named entity recognition (NER) model such as a ⁵⁷bidirectional long short-term memory - conditional random fields (BiLSTM-CRF) model (Lample et al., 2016), which is the initial capacity of the model. Second, much more unlabeled data is kept as the source of information.

After the initial model is trained, the model then infers the unlabeled data and the uncertainty of the model is calculated. Using uncertainty sampling, the samples where the model is least confident are selected. These samples are thought to be the most informative as they are likely to provide the greatest improvement when labelled (Settles, 2012). But using only uncertainty can result in the same sample being chosen more than once. To overcome this redundancy, the method can also use diversity selection, where the samples from various parts of the data space are selected (Sener & Savarese, 2018).

These are then shown to the human annotator, using the NER tagging scheme. The new samples are then incorporated into the training set, and the model is retrained with the new samples. This process continues, creating an iterative training process, where the model can be continually refined, and the number of samples to be labelled is reduced.

The major advantage of this approach is the exploration/exploitation trade-off. Sampling with uncertainty aims to sample difficult cases (exploitation) and sampling for diversity aims to sample diverse cases (exploration). It leads to faster learning than in standard supervised learning methods that adopt random sampling.

One other vital element of the suggested framework is its neutrality and versatility with regards to data and operations environments. In many real-world applications of NLP, for some entity types there might be few number of entities present while for the others there could be a lot of entities. Minaee et al., (2021) noted that supervised learning models are generally under-trained for finding the underrepresented entities during training, which can result in differences between the entities of minority and underrepresented classes. The existing drawback is that, presented samples during the NER are not selected based on the most informative and uncertain but instead on samples picked up at random. Therefore, this model can learn more broadly from a select and discerning subset of the data. In particular for niche areas such as legal text processing, healthcare, finance and other specific regions requiring a high level of expertise and a significant number of people on the part of the annotator to perform this task (Hedderich et al., 2021).

The recommended strategy also allows for scalability and continuous modifications for development. In the world, text is constantly evolving and is never finished, new words, companies, products and social movements arrive all the time. The effectiveness of static NER systems gradually deteriorates over time, due to the changes in language. (Li et al., 2022) During the active learning process, the model can be continually fed the newly selected informative samples for learning, and the system can be updated without a need to re-train it from the beginning. This is especially useful in the context of online communications and businesses in which languages and usage evolve constantly.

The other benefit of the framework is its efficiency in terms of annotations. Manual NER annotation is time consuming and expensive because it requires finding the exact start and end-point of entities and the exact classification of them. The annotation cost is still one of the crucial hurdles in the supervised machine learning. First, in the following, we learn a basic network from fewer annotations for a subset of a large and uncertain input set. As a result, the organisations are able to develop highly effective NLP systems at more affordable cost and at less operational costs.

Additionally, the framework adopts suggestions from human-in-the-loop AI-focused technologies that involve both the machine learning models and human skills to improve energy efficiency and the quality of learning. The model does not fully automate the process of annotation, but rather picks out the ambiguous cases; or it is not just used for leaving hard cases to be annotated by human annotators equipped with good labels, but it is used for picking out hard cases which are equipped with good labels. The process of annotation is not fully automated by the model, but rather, hard cases are identified and high quality labels are made by human annotators for the hard cases. The benefits of human-in-the-loop learning in improving machine learning systems in low resource settings (HILoL) have been gaining ground (Holzinger, 2016). The framework is still a dependable and minimizes cumbersome labeling.

From a methodological perspective, the significant part of the model and the method of queries is designed to be flexible. The transformer based model like BERT, RoBERTa and the domain specific language models can be used as a baseline model, and in this work, a BiLSTM-CRF model is used. Similarly, the main sampling methods used in this study are uncertainty and diversity sampling, and other sampling methods may be used such as a query by committee or density weighted sampling methods (depending on data type and the application's needs) (Settles, 2012). This flexibility allows the framework for the protection of children to be implemented in a better way and to develop it in a methodological manner.

The framework also provides a systematic method for comparing the utility of the different kinds of annotations given different budgets for adding annotations. By conducting its experiments iteratively, one can monitor and evaluate the change of the predictive performance of the model after multi-labeling, thereby determining which samples will be most beneficial for boosting the model's predictive power while incurring the lowest cost of multi-labeling. In an NLP projects, this can be used to help decide at what point the resources are being allocated may not be needed to add much (Sener & Savarese, 2018).

Overall, the proposed conceptual model is a systematic model of a combination of active learning in Named Entity Recognition. The framework tackles important issues related to data sparsity, low label cost and adapts the model in an iterative training loop through combining uncertainty-based sample selection and diversity-based sample selection. The framework is also designed to be efficient in using the learning process and also facilitates increased deployment or the creation of the model in a real world environment. Hence it gives the conceptual base for the methodological implementation that will be the topic for next chapter.

CHAPTER 2. METHODOLOGY: PROPOSED ACTIVE LEARNING FRAMEWORK FOR NER

2.1 Problem Formulation, Dataset Description and Preprocessing

Named Entity Recognition (NER) is one of the notable NLP operations, which entails the identification and classification of entities, including organisations, places, and numeric properties in a non formatted piece of work. Although neural networks architecture has reached a significant degree of achievement and development, especially proving of sequence framework and transformer-based NER systems, the success of NER systems is directly related to big and large-quality labelled datasets. Surely a time-intensive, resource-intensive process is needed to annotate such datasets, and such could not be possible unless the help of the human resource as it can determine the boundaries of the entities, and label them with appropriate label given to them. This dependency is an ultimate weakness when it comes to real world applications particularly in those areas where the labeled data is small or too expensive to acquire (Settles, 2012).

Figure 1. Interactive machine learning with corrective annotation puts the human-in-the-loop during the model training process.

Source: Smith, A. G., Petersen, J., Terrones-Campos, C., Berthelsen, A. K., Forbes, N. J., Darkner, S., Specht, L., & Vogeliu, I. R. (2021). RootPainter3D: Interactive-machine-learning enables rapid and accurate contouring for radiotherapy. *Medical Physics*, 49(1), 461-473.

As a way of resolving this dilemma, this paper shall undertake studies on adoption of active learning as a model of reducing the annotations to achieve competitive models outcome. Active learning is a paradigm where the learning algorithm is selective in its queries of which items to be labelled are the most informative and is no longer dependent on randomly chosen training data to make these queries. The general idea behind such thing is that selected good samples can turn out to be more of the learning value than homogenous selected samples hence improving data economy and minimizing the expenditures of labels (Cohn et al., 1996).

In such setting, the issue that is discussed in the given research can be molded in the manner shown below. Having had a collection of textual descriptions regarding online courses, one would have to say that designing an NER system will be able to process the data of the pertinent type which will include the information about the course type, partner colleges, price data, the value of the course length, course level among others. These are objects that are distributed as semi-structure textual objects (including title of courses and description of courses) and have to be revealed in methods of sequence labeling. This is the issue of entity recognition performance that will be able not only to reach but also not to be annotated with the entire dataset.

The notationally corresponds to and gives large volumes of unlabeled textual information by U and a comparatively small volume of labelled information by L. It aims at building a training set which is optimum in an iterative sense i.e. an optimum training set is built by building on a sample S= Usuch that it would be optimal in the sense of performances of the NER model once built on top of and labelled with L. The amount of labeled samples, which is allowable, is capped and this is commonly known as annotation budget. This issue may therefore be defined as an optimization problem where the focus that has to be maximized is the maximum with minimum labeling cost model.

This paper will employ the active learning scheme which involves a pool-based scheme model wherein all the data that are not labeled are known in hand. Every iteration is accompanied by a retention of the present model at marked set on the labelled data Land, and at the unlabeled pool U in order to reach informatively predeterminations of the individual sample. Query strategy is used to pick the most informative samples which are contained in the training set. This circular reasoning will go on until the budget on annotation runs out or the performance advancement flattens. The structure is especially useful in the context of NLP topics, in which huge volumes of text that cannot be tagged are not a challenge to acquire (Settles, 2012). The important element of this formulation is informativeness definition. Informativeness of the case of this study is strongly tied with uncertainty on the model and assumptions that samples where there are high confidence of the model have the more chances of exhibiting useful learning. But the uncertainty is not the most common criterion which can be helpful since it can result in the choice of redundant or noisy samples. It thus, takes into account the effort of diversity in the selected procedures that the selected samples break the scope of data distribution too. This uncertainty and diversity is to equalize exploration and exploitation to the more efficient way of learning.

The other significant aspect of the issue is its practical aspect. Data sample of this research is the information of the real-life educational platform as the information concerning courses is continuously generated and altered. In this case, any existing data cannot be manually labeled, thus efficient methods of annotation are needed in this case. The given research is focused on proving how machine learning models can be scaled with active learning as the part of NER pipeline that should be taken into account to satisfy the demands of the real world.

The figure illustrates the proposed active learning framework for Named Entity Recognition. The process begins with a small labeled dataset used to train a baseline BiLSTM-CRF model. The model is applied to an unlabeled data pool, where informative samples are selected using a hybrid uncertainty and diversity-based strategy. Selected samples are annotated and incorporated into the training dataset, and the model is iteratively retrained to improve performance while minimizing labeling effort.

Figure 2. Proposed Active Learning Framework for Named Entity Recognition

Source: Created by Author

The online classes are the 1,000 in online course failures which are available on the Future learner site and will be used as sample data in the study. It is a structured and multi-lateral bundle of learning content on 14 areas of knowledge and utilization of more than 100 partner universities. Every dataset record has a certain number of fields which are name of the course, description of the course (text), category, course length (in weeks), course level, course price, ratings of the learners and course enrolment. This generosity of this information gives an appropriate ground to Named Entity Recognition (NER) efforts and additional analysis concerning business content.

Table 2. Dataset Structure Table

Feature	Description
course_title	Name of the course
category	Subject area
price	Course cost
duration_weeks	Course length
enrollment_count	Number of learners
rating	User rating

Source: Created by Author

In the instance of natural language processing, it is semi-structured written information, and these beneficial items are integrated into description fields. More specifically, the opening text of NER model would be the introduction of courses and course descriptions. The courses, institutional attachment to and quantitative properties, all owe to the disciplines and enable them to perform the labeling sequence tasks. It is also the structured features like price, duration and enrollment that are also considered in the analysis, but the problem of NER model is concerned with extracting the entities of textual information of a text.

The language and practice friendly are the subtypes of the entities which have been developed in this study. These are the organizational items (e.g. partner establishments), the categorical items (e.g. study topic) as well as the numeric items (e.g. pricing, course time frame and enrolment numbers). Having the emphasis laid on such entities, hopefully, the research will contribute to bridging the gap between textual information extraction and reality application of analysis. Through the chosen forms of entities created, it is, thus, not solely the established NER rules according to which the type of the entities selected are predetermined, but also the specifics of the source data used and received and the purpose of its utilization.

There is a set of preprocessing that is being done on the dataset to model. The initial level is data cleaning which is with regards to the lack of values in a

dataset or discrepancy of value. In this case as an example, data that has been done to records of unpriced and untested rating data is normalised or sifted to make sure that the data is consistent. This will be required to determine the effectiveness of the NER model, and the analysis. Normalization of textual data is observed to improve model performance, as well as, the variability or inconsistency. Normalization involves text-to-text translation into some standard form, elimination of redundant symbols and expression of mathematical notations in standard numbers. It can be done by the step to make sure that similar phrases are coded similarly hence the model is knowledgeable about them in a more appropriate manner. The next step after normalization is tokenizing after which the text is divided into the smallest unit (tokens) which is the smallest unit in sequence labelling. NLP process requires the tokenization because it determines the granularity level on which the model works at the text level. It appears that all tokens have labels by the standard BIO (Beginning, Inside, Outside) system of tagging and are usually used together with tasks of NER. In the scheme, the limits and the nature of elements in a sequence may be determined through the model (Tjong Kim Sang & De Meulder, 2003). One of the most significant pipelines of preprocessing is production of original labeled dataset. As the research is a learning model of the active type, few data are initially annotated. This is selected randomly as such seed dataset and further on it is used to give a valid starting point that would be used to train a model. All of the chosen samples will be marked on the token level and one can make sure that the limits and designations of elements are established appropriately. The rest of the data is taken as an unclassified pool that undergoes the sampling process step in the active learning. It is also complemented by the token-level labeling, but there also is a stage of preprocessing that implies the re-remaking of data in the format that could be utilized by the selected NER model. It involves sorting of tokens and labels to sequences and matching them against the input features e.g. word embeddings. The representations help the model to be trained on lexical as well as contextual information. The percentage of the various entities is also another measure that should be considered when pre-processing. In real life databases the entities may consist of class imbalance with a few of them being very prevalent as compared to the others. This has not been explicitly resampling or reweighting this study despite the fact that the active learning framework ignores this problem in the sense that it inherently biases the focus on the informative samples which can contain underrepresented classes of entities. This will help in greater balance and successful learning process. Lastly, preprocessing pipeline will be used to make sure the data is partitioned into the corresponding subsets that will be utilized in the training, validation and testing. Although the major emphasis is placed on the so called labeled and unlabeled split as the participants of an active learning process, the other validation mechanisms are to check on the model performance and avoid overfitting. This is a methodological manner of data preparation contributing to the sustenance of technical objectives of the study and reproducibility of the experiment. Concisely, the data applied in this research is a prolific and diverse basis of NER and active learning. This data is converted to a type of data which can be used to carry out sequence labeling tasks by doing a small amount of pre-processing (e.g. cleaning, normalization, tokenization and annotation). They are also included, with the integration of the textual and structural factors adding to the analytical depth of the information since, in this manner, the study would be capable of matching machine learning facilities with the corresponding business expertise.

2.2 Baseline NER Model Architecture

To measure the suitability of the given active learning framework, a baseline Named Entity Recognition (NER) model will be built through a supervised manner of learning. Such data model is geared towards providing a level of reference upon which performance would be measured against the point of reference in a manner that the impact of active as a learning process would be well estimated in a known environment. The model architecture adopted in the present work is that of a Bidirectional Long Short-Memory ⁶network archived with a Conditional Random Field ⁶ layer that is widely known as an efficient and effective kind of a model when it comes to sequence labeling tasks are considered (Lample et al., 2016).

Figure 3. Architecture of BiLSTM-CRF model

Source: ³¹Ji, B., Liu, R., Li, S., Yu, J., Wu, Q., Tan, Y., & Wu, J. (2019). A hybrid approach for named entity recognition in Chinese electronic medical record. ⁴BMC Medical Informatics and Decision Making, 19(S2), 64. <https://doi.org/10.1186/s12911-019-0767-2>

The preference of the architecture has been made due to the fact the BiLSTM-CRF architecture allows focusing on both contextual and structural dependencies between texts and labels. It plays a very crucial role whenever classifying a given token so as to put a given token into context in case of NER tasks. A good example is that of interpreting a word as a place or organization which may shift with taking into account the word to the neighbour words. The high performance therefore should have the models that can bring the contextual information on it.

The main element of the model is ¹¹bidirectional Long Short-Term Memory (BiLSTM) network. ¹¹Recurrent neural network It is presumed that ¹¹Long Short-Term Memory (LSTM) networks will solve these issues in terms of the traditional RNN, especially, the vanishing gradient issue. With the help of gating processes which regulate information flow that helps the model to learn long range dependencies between sequences is learned (Hochreiter and Schmidhuber, 1997). The significance of this ability in the scenario of NER is gigantic, as the details about the scenario can be represented as many tokens distant in comparison to the entity under classification.

This is also expanded to the bi-directional capability of the LSTM that has the ability of performing in both ways in the forward and reverse direction on the input sequence. It implies that the model will be capable of getting certain context, both the past and the future, of each individual token. Subsequently, a connotation of each of the tokens would be more detailed due to a representation of all of the texts surrounding. This two way processing is demonstrated to be considerably performance oracle on a sequence labeling task, i.e. NER (Graves and Schmidhuber, 2005).

Word embeddings are word vectors, and represent input tokens as denser high-dimensional vectors at an ensemble space. It is these embeddings that develop semantic and syntactic word-word relationships, and through which the model is able to generalize more in the similar situations. The embedding initials are not pretrained and sampled randomly and trained in training in this study; or, in situations where this is not possible because of implementation constraints, the trained source is made use of. The embeddings also make it possible to do without operation feature engineering, and make the model itself learn to treat untreated text only.

The contextual representations of the whole sequence of tokens are the outputs of the BiLSTM layer. These representations are in turn fed to a Conditional Random Field (CRF) layer that is the decode side of the model. CRF layer has the responsibility of prediction of the most preferable sequence of labels with regard to dependence among the adjoining labels. This is further more in NER where some of the labels tendencies are not legal in the BIO tagging scheme. Since, it can not be found that there is I (Inside) tag, there is a O (Outside) tag and that there is no B (Beginning) tag between I and O.

The layer is referred to as the CRF that is capable of modelling the conditional probability of a complete sequence of labels and hence ensure that there is consistency of global prediction and proper transition constraints (Lafferty et al., 2001). This also leads to a better and consistent body recognition compared to the models which track tokens individually.

The training process of a baseline model is then optimized by the model optimization of parameters of the components BiLSTM and CRF with the assistance of both the label information. The model will be trained on estimating the input sequences so as to label the sequences through learning a loss function equal to the negative log-likelihood of label sequence correct. They are optimized by a gradient and trained till a convergence point or they have

reached some degree of performance on a validation set.

2.3 Active Learning Strategy Design

In this study, uncertainty is measured using the prediction confidence of the model based on the probability distribution over possible labels. Samples with lower maximum probability scores are considered more uncertain and therefore more informative for annotation. At the same time, diversity is approximated by selecting samples that are less similar to each other in terms of their contextual representations, ensuring that different parts of the data distribution are covered. During each iteration, the model selects a batch of samples that have the highest uncertainty while maintaining diversity, and these samples are then sent for annotation and added to the training set.

The success of the proposed program is very dependent on the strategy of active learning that will specify the way of choosing informative samples of the unlabeled information. Active learning as compared to the traditional supervised learning where the training data are chosen randomly, although this method focuses on the samples which can be learned best by ranking the samples, based on their contribution to the improvement of the model. The proposed study encompasses an active learning paradigm of a pool that is specifically effective in Natural Language Processing (NLP) assignments because the pool will always feature an enormous community of unmarked text (Settles, 2012).

The active learning concept is founded on the concept of informativeness that is expected value of a sample of data of enhancing the performance of a model. Uncertainty of models is the main measure of informative in this study. This is because the samples and hence the model lacks confidence in its prediction but rather predicts which have more chances of providing practical information that can be used to establish the boundaries of decision of the model. It is a hypothesis that has been extensively well confirmed in the rank of active learning hierarchies and that is prototypical with most of the pragmatic query schemes (Lewis and Gale, 1994).

Probability distributions assist in the estimation of uncertainty resulting in NER model. The model allocated every designation on the likelihood of every entity within a sequence. They can be aggregated and used to establish the confidence of the model as a whole on its outputs in the sequence level. The lower confidence samples will be rated higher and hence preferred in their annotation. This will be applicable in sequence labelling to make sure that a part of a sentence that has the ambiguous or challenging object is chosen and the context surrounding this object does matter in the labeling activity. Farthing addition is meant to make sure that the sample selected would be representative of other parts of data distribution. This is especially when the content information is a heterogeneous one like in this instance during this study as it represents a combination of subjects and institutions. This assists in creating variety since the model is exposed to additional lingual patterns and varieties of things thereby enhancing the generalization of such patterns and varieties. This can, in practice, be approximated by getting the samples of that candidate which are less similar to the previously sampled samples (Sener and Savarese, 2018).

This is the design element that is important in the process because it is the batch size, which connotes the number of samples which must be chosen every time. It is also not able to get slow learning using the smaller sample number and also give a massive number of samples leading to the generation of less useful query strategy because of a handful of less informative immediate. This paper employs the balanced batch size to make sure that besides the efficient learning, the quality of the chosen samples will also be retained.

The other important attributes of the strategy are the stopping criterion. When the annotation budget has been achieved or when the performance gains in the iterations are insignificant, then, the active learning process is stopped. This guarantees it the successful utilization of the resources and avoids the unsuccessful process of annotation in cases where the model has already reached the satisfactory level of performance.

2.4 Annotation Process and Iterative Training Workflow

The action component of the proposed active learning model is the annotation component and the training part that is iterative. The theories of active learning through such components are converted to a practical extent that promotes training models in an effective manner in a less labelling context. The Named Entity Recognition (NER) meaning of this process is especially resource-intensive due to the labelling of the tokens and providing the right demarcation of where the objects are located. This is why, the research on the development of a successful workflow in the process of annotation might be touched upon as the key to the obtained objectives of the current research.

The active learning process follows an iterative loop as shown in Figure 3. At each iteration, the model is trained on the labeled dataset and used to evaluate the unlabeled pool. Based on the query strategy, informative samples are selected, annotated, and added to the training set. This process continues until the annotation budget is reached or performance stabilizes.

The samples are then manually labeled to the labelled dataset with the mark on them. It can be applied to selectively expand the training data to ensure that even new examples of labeling provide the model with useful information. When this time the labeled dataset is patented with the enlarged data, the model is again trained with the enlarged, evidently, enlarged data. It is a re-training process that will cause the model to embody new knowledge, and will cause it to become flawless predictions.

Figure 4. Active Learning Workflow

Source: Created by Author

Annotation process tries to start with the production of an initial labeled set otherwise known as seed set. The data set is constructed using a small sample of the whole sample of the dataset, and is then manually randomly marked in accordance with some fixed rules. The seed set is executed in order to take the original learning data needed to generate an initial NER model. Nevertheless, even the small seed dataset is supposed to be representative such that it enables the model to be sensitised towards the simple tendencies, so that it can yield predictions in unseen information that has a meaning.

Table 3. Active Learning Iteration Table

Iteration	Labeled Samples	Selected Samples	Model Update
1	100	50	Retrained
2	150	50	Retrained

Source: Created by Author

The annotation is represented under the BIO tagging scheme, and the tokens in the scheme have a mark of where there is an entity as the beginning (B), inside (I) or outside (O). This scheme allows the model to bring about entity boundaries and types in an organized way. Considering the type of data, annotations are spent upon the identification of such objects as category of courses, the institutions which have collaborated with the course, price, duration, and other types of variables in textual situations. The rationale is that annotation rules are formalized to guarantee that all things are identical, the way the different respectively types of objects will be listed, and how a case that arises because of some ambiguity will be handled. The annotations consistency is absolute because inconsistency at any point can have a negative influence on the outputs of the models that are predicting results of inconsistency.

One again repeats the active learning process based on the initial model training on the seed data. Each iteration of the trained model is used on prediction and estimation of the informativeness of all samples in the unlabeled dataset pool. Active learning strategy introduced in the previous section forms the basis in the annotation of a selected sample. This is the reason why these samples are chosen since they will offer the highest utility in modelling the performance since these samples are the most uncertain samples or simply because there is some diversity in the data distribution part. The retraining cycle selection-annotation-selection is repeated to form a cyclic chain of actions and is practically an enclosed envisage of training feedback. It is factual to affirm that the model will increasingly become more accurate with each step because it relies on a data that is more informative. It is a cyclical procedure and enables a single piece of work to be refined on a continuous basis whilst checking the expenditures of annotation. The annotation budgeting is one of the substantial elements of the provided workflow. In practice, there are limited data labeling resources abilities but, therefore, it has to have a limiting quantity of samples which might be labeled. The paper makes a simulated annotation budget where it is assumed that there is a given budget. The budget specifies the overall amount of the samples that may be labelled during the active learning, and such decision variables include the batch size, and halting parameters. The sample size of a batch is the number of samples that is picked of each iteration. The process of a good batch will be related to efficiency or effectiveness. Larger batch sizes may accelerate the rate of buying data but with the larger selectiveness of such samples at the expense of smaller ones being capable of refining the model itself and it refines the process. The presented study assumes a moderate batch size, as a consequence of which, following every round, they obtain substantial gains that are independent of the alterations in the quality of the sampled samples. The other factor that should be considered is the termination factor of active learning process. The workflow is terminated when the budget of the annotation is exhausted or the discrepancy between the outcomes of one instance and the other is minimal. The inter-iteration performance then must be monitored with the intention of determining when there is nothing more that can assist in the annotation. It is normally performed by testing a validation sample and assessing the model using the measures of precision, recall and F1-score. Learning curves which can be seen as the dependence between model performance and a controlled number of samples that are labelled can also be built using iterative training workflow. The curves make it possible to take the perspective of the efficiency of the active learning process and demonstrate the rate on which the model upgrades and constructs the points of diminishing returns. It is needed in the analysis of the trade-off that was present in effort in annotation and predictive performance.

CHAPTER 3. FINDINGS AND DISCUSSION

3.1 Dataset Overview and Experimental Setup

Online courses that are gathered in FutureLearn platform with 14 clusters of topics and over 100 partner organizations in total are included in the database used in this study (1,000 courses are found online). There are free and paid courses included in the dataset, as the former constitute about 87% of the total catalogue, and on the other hand, paid courses 13%. Different courses such as price, and course length (1-11 weeks) vary by an increment of 0-120 pounds. The learners have an average mean score of 4.70 on the overall platform compared to 5.0 which means that overall, the learners are satisfied with the platform. Each course data is categorized into the following e.g. categories, cost, time, level of difficulty, partner school, the number of people enrolled and rating of a student. These features not only facilitate technically measuring the performance of the NER model, but the performance of the business oriented assessment platform. This information is organized and grouped of the items that have been extracted of the NER system and then analysed further.

Table 4. Platform Snapshot

Metric	Value
Total courses analysed	1,000
Subject categories	14
Unique partner institutions	100+
Free courses	870 (87%)
Paid courses	130 (13%)
Avg learner rating	4.70 / 5.0
Price range	£0 - £120
Course durations	1 - 11 weeks

Source: Created by Author

The comparison between the performance of the baseline model and the proposed model of active learning and limited labeling conditions are the elements the experimental setting will be grounded on. Besides the evaluation of the models, some extra observations of the structured data is provided to recognize the patterns of demands among the learners, pricing policies, course design and performance of the partners. The two-way approach also contributes to the research in that it enables the research to include both practice and technical uses of platforms optimization. The diversity of the dataset also means that a good sample of the various types of education which are available online was recorded and multiple variables of student activity and platform effectiveness were measured. Massive areas of understanding, pricing tiers and course lengths provide the ability to evaluate the impact of regular textual information in machine studying and business analysis. In particular, the data is actual data from a real environment of online learning - some courses are very popular; some are very unpopular; some students can study some but not others; some courses are important to some but not others; some courses are valuable to some but not others. Several preprocessing steps were taken in order to attain consistent and usable data prior to the creation of the implementation. The data used to create the implementation were preprocessed in a number of steps for consistency and usability. All entries that included text data (course titles and descriptions) were cleaned, such as the removal of extraneous symbols, repeats and incomplete items. The text was then tokenized and tagged using the BIO labelling approach to be used for Named entity Recognition tasks. Systematically, a variety of these entities, such as Category, Educational Attribute, Pricing Related etc. have been identified and extracted from the text data. Another property of the data set is that of the imbalance between paid courses and free courses. Very few courses in a catalogue of some sort (which would be used in the actual data source if the data was to be obtained via this dataset) were paid for and the most of them on digital platforms were accessibility-oriented courses. The fact that these highly-enrolled paid courses do exist, at the same time, provides an opportunity to examine the relationship between pricing strategy and student enrollments. The NER dataset is thus not only acceptable for technical analysis of the NER framework, but for general analysis of platform optimization and usage patterns as well. The experimental design also lets the observation of model behaviour with restricted annotations conditions. The active learning framework carefully chooses the samples to be annotated over time based on some data which is already partially labelled, and thus uses less labelled data for effective

learning. This setup is similar to the actual practice done in the NLP community, in which large-scale annotation may not be economically feasible and/or practical in terms of operations. Thus, the data set could be used for evaluation of the efficiency of the annotation as well as suitability of active learning method in real-world applications.

3.2 Model Performance and Learning Efficiency

Standard measures such as ⁴⁶precision, recall and F1-score ⁴⁶test parameters were considered to evaluate the performance of the baseline NER model. A stable point where the proposed active learning approach was compared is the successfully trained full-label baseline model that is trained on a fully labelled dataset.

Active learning model: The active learning model was sequentially experimented on the annotation steps on the various sections of the unlabeled data whereby, a portion of the unlabeled data was randomly selected and annotated according to the query strategy. The results indicate that the increases in the performance of the model are slow at each iteration meaning that the selected samples are informative and are beneficial in learning process. Some of the important findings obtained include the fact that the active learning model can be trained to be not worse than fully supervised baseline using much fewer rated samples. The high improvement rate exists with the improvement in the initial models which might make the performance an advantage of using high uncertainty and nonhomogeneous samples. The more instances annotated, the higher the performance, indicative of diminishing significance in adding more annotation.

In terms of efficiency of the annotation process, the offered solution enables to decrease the number of labeled data needed to achieve some degree of performance. This indicates the effectiveness of active learning strategy to minimise the labeling effort with the maintenance of model accuracy. The findings present here are a confirmation that selective sampling is a more superior approach to use as opposed to random annotation particularly when the training process is in its early stages.

Overall, the findings demonstrate that active learning provides a practical and efficient solution for Named Entity Recognition tasks in scenarios where labeled data is limited or costly to obtain.

Results also indicate how suitable active learning is to get better performance with less effort in learning, as it concentrates on informative samples instead of annotating them comprehensively. In the initial versions of the training protocol, it was able to attain significant performance improvement even with a small fraction of the labeled data. That is, with good sampling the samples are more useful in improving a model than randomly selected samples. Specifically, the system learned to sample more ambiguously and more challenging instances by the uncertainty based sampling and ensured to sample more diversely in terms of regions in the dataset by the diversity based sampling. These strategies enabled the model's ability to be more effective learning relevant entity patterns trainings at the beginning.

An additional important finding is that even with the widely varying amount of labeled data used there is still a relatively low difference between the performance of the baseline and the active learning model. The fully supervised algorithm on the full labeled data set was able to achieve an F1 score of 0.89; the active learning framework achieved an F1 score of 0.87 for the 40% labeled data subset of the full data set. This underscores the role of active learning in reducing the need for data marking, and its importance in maintaining competitive prediction accuracy. In an operational context, this labelling effort is a very significant savings in considering the costs of preparing datasets, and of speeding up the time taken to produce a dataset required for a real-world NLP project.

The learning curve also implies that after a few iterations of annotations the improvement in the model tends to converge. The more samples are put into the training set the less there will be in the data set that have informative labels. Consequently, active learning may result in increased annotation effectiveness, if not even into the definition of an optimal trade-off of annotation performance and cost. Results indicate that this could be a scalable option to currently adopted fully supervised pipelines in resource limited settings where a majority of the NLP models have been trained, because often it is limited labeled data that is available in these settings.

3.3 Business-Oriented Analysis of Extracted Data

Figure 5. Category Overview: Course Volume vs Learner Demand

Source: Created by Author

The demand and supply of the courses among the learners has significant differences with the categories, which shows that the platform has structural discrepancies. The most common course (216 courses make up 21.6 percent of the dataset) but has lowest average course enrolment at an average of 10,981 learners making it the least efficient course in relation to the entire dataset.

On the contrary, the strongest courses that are in demand are the Language courses, the average number of Enrolment in the course is 34, 673 learners per course however, there is only 58 courses available in the Language courses. This implies that there is gross under-supply in high need segment. Similarly, the Study Skills and History have good scores, because they have average scores of 14,195 and 13,381 in enrollments, respectively, as compared to bigger categories such as Healthcare and Medicine (9,132).

Healthcare & Medicine is a high volume category, and is continually demanding with a consistent performance, although there is no streamlining. On the other hand, one has to mention that IT and Computer science despite the number of courses mentioning 113 courses only in total have a very low average of about 5,153 that is, inefficiency of their supply in terms of the activity of the learners.

Such tips would imply that category expansion strategies need to focus on an expansion of high demand categories like the unrepresented ones like the Language and refine rather than adopt categories which are already over-represented.

Figure 6. Free vs Paid Courses: Enrollment & Revenue Opportunity

Source: Created by Author

In contrast, free courses and paid courses have a marked difference in the sense of engagement of learners. The number of learners enrolling in paid courses is 46 averagely and 291 averagely, as compared to the free courses with average 3,049 learners enrolling in each course. This is a difference in the degree of engagement about 15 times.

The hole in structure is further compounded by the scatter of enrollments. The vast majority of the free courses have been at the lower enrolments scales (smaller than 5 000 learners) but the high cost courses at the high level especially those courses in excess of 20 000 learners. This reveals that the difference as seen is not influenced by the outliers but there is an overall trend that has been prominent in the dataset.

This finding was a quite contradictory one with the traditional belief that the optimal engagement would be with free contents. Instead it is the perceived value, quality of production and dedication to the course which could be good in paid courses.

Strategically, it seems that free courses have a more of an entry level product, and the value of activities and platform is paid courses.

Figure 7. Price Point Strategy: Volume & Learner Response

Source: Created by Author

Price point formations are also indicative of particular tendencies on the knowing part of the learners. The most frequent price level is the £39 price level, which has 55 chures, and has a good enrolment performance. It should be noted that the higher level (59) average enrollment is lower than the lower (39) and higher (79) levels average enrollment of approximately 64820 (higher engagement also at that level, 79) is, with the lower level with a smaller sample size (at 39).

This implies that the mid-range pricing might not be appropriate in narrating the value thereby less participation of the learners. Conversely, higher quality cue, which is perceived by the higher price level like the £79 appear and undercut accessibility, as would occur with the lower price level.

The upper-end (99120) is underrepresented, yet points to the fact the high-end location will as well suit niche groups of learners.

In general, the findings suggest that pricing does not need to play a confounding middle ground positioning, but rather, certain values.

Figure 8. Optimal Course Length: Duration vs Learner Engagement

Source: Created by Author

The length of the course determines student engagement. The highest enrolments lie at course 5 weeks and the average of 5 weeks courses is 15343 students or enough content and time.

The less popular courses are the shorter courses (e.g. format 1 week), with the number of learners being very low (around 1,564) and that can be considered undeserving. Meanwhile, the levels of engagement of the longer courses are also declining, and the participation of 8-week courses has an average of 1,029 learners, one of the lowest involvement rates.

The optimum model of performance (13,700 learning) is the middle-range courses, which include the most frequent model of course delivery i.e. 2-week courses i.e. a staple course delivery model.

These findings show that there exists a range of optimal period (approximately 2-5 weeks), beyond which the investment in participation by learners reduces since this is the period during which a greater amount of time investment is implicated and the risk of dropout arises.

Figure 9. Partner Performance: Total Reach vs Per-Course Efficiency

Source: Created by Author

There is a high variance in total reach, and efficiency per course at the partner level. This happens to be the University of Leeds with the highest total enrollment of all such i.e. about 2.27 million total enrollments due to the large number of courses being offered. Its average course enrolled (24,135) is however average when compared to other partners which are more efficient.

University of Groningen on the contrary has the best performance with the performance mean of 80,435 students in the courses and few courses.

Likewise, Vrije Universiteit Amsterdam (68,748) and University of Reading (51 253) have good per-course performance.

However, there are alternatives partners that do contain large number of portfolios of an academic nature course (e.g., Taipei Medical University 54 courses) with fairly small average course enrolments (~4,116) and though course is rich inefficiencies of content performance can be identified.

The implications of the results would be that the two strategies i.e. scale based reach and efficiency based impact are achieved. The partners which can be pursued as expansion partners include the high efficiency partners and the poor performing high-volume partners should be optimised on content.

Figure 10. Strategic Opportunity Map: Satisfaction vs Learner Demand (Bubble size=number of courses offered)

Source: Created by Author

The strategic opportunity map will combine both needs and satisfaction of the learners with the intention of allowing these learners to have a multi dimensional analysis of the category performance. Based on median values of demand and satisfaction, the chart is divided into four quadrants. Top-right segment categories are denoted as star performers and are a combination of high demand with high satisfaction. This obviously becomes overtaken by the language whose average enrollment (~34,673) is the highest and level of satisfaction is among the highest (~4.76). This means that the market has a needs with high level of demand and quality delivery.

Such categories as Healthcare and Medicine and Psychology and Mental Health are used instead of the high-satisfaction, moderate-demand category and indicate the high quality of the content and a probable growth. IT and Computer science, on the other hand, lies in the less-demand segment having a moderate (~4.68) degree of satisfaction that suggests that the field has inefficiencies in attracting the learners compared to its size.

The category of lower right quadrant (Science, Engineering and Maths), (approximately 4.55 rating, approximately 6,775 learners) are probably the danger zones since it is not only highly demanded, but it is also rather unsatisfied.

The point-to-point emphasis on these strategic choices offered by this model can isolate the prospects of growth, the demand of quality additions and the unprofitable segments.

3.4 Discussion and Summary of Key Findings

Based on the findings in the section above, a number of persistent trends is identified, including the behavioral and performance of learners, and their strategic placement on FutureLearn platform. In this case short results of these are added and analytical and practical implication of these results are given.

Some of the primary results of the analysis that deserve mention include the fact that there are some categorical supply and demand mismatches in both directions. The number of courses offered by such disciplines like Business and Management is most prevalent (216 course) but the average rate of enrolment in the courses (~ 10,981) are not the most preferable degrees that the learner can study. On the other hand, courses belonging to the Language category are enrolled in very high (average of 34, 673 students per course) but with a comparatively lesser number of courses (58 courses). This skew indicates that the growth of platforms does not need to be capped by the cumulative content, but an allocation of resources across the world by category. This would be a strategic move as the portfolio would have been rebalanced and the proportion of the high demand and under supply one would be the most profitable than just to inject more courses in the portfolio.

The other important outcome is connected with the influence of pricing on the development of the learner behavior. A comparison of the two reveals that the paid classes have much better possibilities of success compared to the free classes (an average of 46,291 and 3,049 learners enrolled, respectively). This fact contradicts the traditional beliefs regarding the participation of users in the online learning environment where open access must probably optimize participation. Instead, this is what results are telling us, that not only is price a deterrent but is promise of quality and investment that leads to higher rates of interaction among paying learners. The pricing and tiers analysis, also reveal that the medium pricing (e.g. £59) is not as effective in comparison to lower (e.g. £39) and higher (e.g. £79) pricing strategies, which can be a sign that the unambiguous positioning (i.e. it is affordable or it is a premium) is effective against the unclear or undefined mid-range pricing strategies.

Learner engagement also highly depends on design characteristics of courses. The data indicate that the course duration is not linearly related to the

enrollments with the highest course duration of 5 weeks with the highest average percentage of engagement (around 15,343 learners) and course duration of very low (1-week) and very high (8-weeks). This implies that there has to be an optimum depth, neither too deep nor too shallow. Short courses can be regarded as valueless yet courses which are excessively long can be regarded as a big set of time demands and, therefore, less attendance. Thus, optimization of course design will be one of the main levers to enhance the performance of the platform without the need to raise the amount of content.

The analogy of the analysis on the level of partners shows the distinction between the model of the performance that is performance-driven by scale, and the model of performance-driven by efficiency. The large institutional access and reach, with a large catalogue base such as at the University of Leeds, are still high total reach, but with high efficiency with an average of over 80,000 learners per course, such as the University of Groningen. It shows that an increase in co-operation with high-performance suppliers can result in returns that are big, but disproportionately, and that large but non-performing catalogues ought to be met with quality improvement. The other reason that continuous performance measurement and content assessment is necessary is that we have high volume and low engagement partners and must be considered at all times.

The strategic opportunity mapping framework that incorporated both the demand and satisfaction as one construction also supports these insights. The ones such as the Language have a dominant role and are high in demand and high satisfaction hence one of the most important strategic assets. In their turn, the categories, in which the level of demand is moderate or low, such as in the framework of acceptable rates, show the areas, in which the content discoverability, relevance or positioning can be improved. Such an approach based on multidimensions would enable perceiving the performance of the platforms at a more subtle level than individual measures would.

Table 5. Executive Decision Summary

Priority	Finding	Recommended Action
1	Language courses exhibit the highest demand but limited supply	Commission 20-30 additional Language courses in collaboration with high-performing partners
2	Paid courses achieve approximately 15× higher enrollment than free courses	Evaluate top-performing free courses for conversion into paid offerings
3	A large proportion of courses lack learner ratings (~87%)	Implement a systematic learner feedback and review collection mechanism
4	University of Groningen demonstrates the highest efficiency (~80,000 learners per course)	Expand collaboration and increase course offerings with high-efficiency partners
5	The £59 pricing tier underperforms relative to £39 and £79	Reposition mid-tier pricing toward clearer value-based tiers
6	5-week courses achieve the highest engagement levels	Prioritize development of courses within the 2-5 week duration range
7	Advanced courses are priced similarly to introductory courses	Introduce differentiated pricing for advanced-level content (£79-£99 range)
8	IT & Computer Science shows high supply but relatively low demand	Conduct content review and refocus on high-demand subdomains
9	Some partners maintain large catalogues with low average engagement	Perform quality and relevance assessment before expanding partnerships
10	A small number of courses generate disproportionately high enrollment	Maintain and extend high-performing courses through updates and advanced modules

Source: Created by Author

Table 6. Interpretation of Priority Levels

Indicator	Meaning
Critical	Immediate action required due to direct impact on engagement or revenue
High	Strategic initiatives to be implemented within the medium term (1-2 quarters)
Medium	Important improvements for long-term platform sustainability
Ongoing	Continuous optimization and maintenance activities

Source: Created by Author

Conclusion

The study has discussed the value that active learning brings towards reducing the labelled data required to attain Named Entity Recognition (NER) and which does not negatively affect the performance of competitive models, and how the structured data the active learning produced can be used to analyse business interests. The results demonstrate that active learning does scale up labeling active samples (with prioritization) to a significant level. This model performance was as well compared to a baseline which was fully supervised and had much less instances to be labelled specifically in an annotation-in-baseline which gave the highest yields. This proceeds to confirm the big assumption that the selective strategies of annotation could be implemented in lieu of labelling exhaustively with practically no predictive power sacrificed.

Besides making methodological contributions, the study offered practical information with the help of an organized study of platform data. The findings provided clear in breakdowns in the supply and demand sides as areas of high demand were underreported and were with the glutton types of demand were overrepresented. Price analysis showed that paid courses had consistently been more successful in their enrolment compared to the free courses, proving the price to be indicative of perceived value, rather than a dis-incentive. Course length analysis also indicated that there is the best course length (around 25 weeks) that will see a higher involvement of the learners as compared to the level-of-partner analysis that showed that efficiency (mean of course enrollments) is valued more than course output. These results demonstrate the importance of strategic suitability of content, price, and framework that contribute to the usefulness of the platform.

Though these contributions have been made, one has to admit a few limitations. The research is based on only one dataset and controlled experimental design that can be a limitation to extrapolating the research to other regions and actually implementation. The NER model looks at the normal types of entities without taking into account the complex architecture, e.g., transformer-based models, which dynamics can be inconsistent in terms of performance. Also, business analysis is conducted with summarized statistics, disregarding long-term changes, and long-term activity of engagement and external factors that can impact the engagement.

Future study may add to the current work as it might be possible to learn actively by more sophisticated models of the NER and apply experimental hybrid approaches i.e. active and semi-supervised learning. It can also be improved with the multilingual and cross-domain data analysis to improve the strength, but user-level analytics might assist in getting a better idea about learners and decision-making approaches.

This study illustrates overall that active learning may as well be a successful and practical method of refining the data annotation in NER and can also

produce conclusions that may be valuable to the business. Knowing the usefulness of a machine-based learning and data-based decision-making, the study would allow to comprehend the problems of the academic knowledge and impact real practice in the real digital platform.

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